ANSWER KEY & MARKING SCHEME SET 1

Chapter 8: **Water Resources (Geography)** | Class 10 Social Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** Meghalaya	200-year-old
2	**(a)** flowing water that restrains and stores it	Controls flow
3	**(a)** Mahanadi	Odisha
4	**(c)** a small share	Accessible water
5	**(d)** Siri Fort area	13th century
6	**(b)** reduces the water storing capacity	Reservoir
7	**(d)** rooftop rainwater harvesting	Karnataka, Mysuru
8	**(a)** rooftop rainwater harvesting	Meghalaya
9	**(a)** Bhagirathi	Uttarakhand
10	**(b)** Sutlej	Punjab-Himachal

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	The hydrological cycle is the continuous circulation of water between oceans, atmosphere and land (1); It includes evaporation, condensation, precipitation and run-off (1).	1+1
12	The Narmada Bachao Andolan is a movement opposing the Sardar Sarovar Dam on the Narmada (1); It was led by local people and activists against displacement and environmental damage (1).	1+1
13	Water demand is rising with population, irrigation, industry and urban use (1); Over-exploitation, pollution and unequal access reduce usable water (1).	1+1
14	Tankas are underground tanks built in houses to store rainwater from roofs, in Bikaner, Phalodi and Barmer (1); The water is used for drinking in dry months (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Sringaverapura near Allahabad had a water harvesting system to channel Ganga floodwater (1); During Chandragupta Maurya's time dams, lakes and irrigation systems were built (1); Bhopal Lake, in the eleventh century, and the Hauz Khas tank, in the fourteenth century, were important tanks (1).	1+1+1
16	Dams may not control floods when heavy rain fills reservoirs (1); Water released suddenly can flood downstream areas (1); Sedimentation reduces the dam's capacity to hold floodwater (1).	1+1+1
17	About 71 per cent of the Earth's surface is covered with water (1); Of this about 96.5 per cent is in oceans (1); Only about 2.5 per cent is fresh water, mostly locked in ice and glaciers (1).	1+1+1
	OR: Industries need large amounts of water (1); Many industries pollute water bodies with waste (1); Rapid industrialisation after independence increased competition for water (1).	1+1+1
18	Growing population increases demand for water (1); Over-exploitation of groundwater for irrigation and industry lowers supply (1); Unequal access and pollution make water scarce in India (1).	1+1+1
	OR: In Shillong, rooftop rainwater harvesting is practised (1); In Meghalaya the 200-year-old bamboo drip irrigation system channels spring water (1); Bamboo pipes deliver water drop by drop to plants (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Rainwater harvesting is collecting rain for later use (1); It recharges groundwater (1); It reduces dependence on dams and tube wells (1); It suits dry regions (1); It is cheap and can be done by households and communities (1).	1+1+1+1+1
	OR: Water is essential for life, farming and industry (1); Only a small share is fresh and usable (1); It is unevenly distributed (1); It can be polluted and over-used (1); So it needs careful use and conservation (1).	1+1+1+1+1

20	Multipurpose projects store water behind dams for many uses (1); They irrigate fields and supply drinking water (1); They generate hydroelectricity (1); They help in flood control (1); Examples are Bhakra-Nangal on the Sutlej and Hirakud on the Mahanadi (1).	1+1+1+1+1
	OR: Dams displace people and submerge villages and forests (1); Silt reduces the reservoir's capacity (1); They change river flow and affect fish migration (1); They may cause floods downstream and conflict between states (1); Costs often outweigh benefits for those who lose land (1).	1+1+1+1+1

Mistakes to avoid:

- Saying water scarcity is caused only by **low rainfall**; over-use, unequal access, pollution and population growth also cause it.
- Matching projects to the wrong rivers: **Hirakud** (Mahanadi), **Bhakra-Nangal** (Sutlej-Beas), **Sardar Sarovar** (Narmada).
- Writing that Mahatma Gandhi called dams the 'temples of modern India'; the phrase is **Jawaharlal Nehru's**.
- Describing multi-purpose projects as only for **irrigation**; they also give power, flood control, water supply and navigation.
- Mixing traditional systems: **tankas** (Rajasthan), **kuls/guls** (Western Himalayas), **bamboo drip** (Meghalaya), **inundation channels** (Bengal flood plains).
- Forgetting that dams have **disadvantages** as well: sedimentation, displacement and damage to aquatic habitats.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 2

Chapter 8: **Water Resources (Geography)** | Class 10 Social Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** renewable resource	Hydrological cycle
2	**(b)** rapid industrialisation and growing demand	Demand
3	**(d)** Cholas	Tamil Nadu
4	**(b)** displacement of people	Social cost
5	**(a)** Western Himalayas	Diversion channels
6	**(b)** 4%	Yet large population
7	**(a)** 2.5%	Rest is saline
8	**(a)** water harvesting structures	Traditional
9	**(c)** continuous circulation of water between the oceans, atmosphere and land	Water cycle
10	**(a)** over-exploitation, rising demand and unequal access	Pressure

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Jawaharlal Nehru believed that dams would lead to agricultural and industrial growth (1); They would be symbols of progress that remove poverty, so he called them 'temples of modern India' (1).	1+1
12	It is a 200-year-old system in Meghalaya of channelling spring water through bamboo pipes to plants (1); Water is delivered drop by drop, saving water (1).	1+1
13	Khadins are traditional water-harvesting structures in Rajasthan (1); They hold run-off on farmland and store moisture for crops (1).	1+1
14	Rainwater harvesting is collecting and storing rainwater for later use (1); It recharges groundwater and reduces dependence on distant sources (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Dams provide water for irrigation (1); They generate hydroelectricity (1); They help in flood control and domestic water supply (1).	1+1+1
16	Gendathur is a village in Mysuru district, Karnataka (1); Households have rooftop harvesting units (1); Roof water is collected in drums and used for drinking and domestic use (1).	1+1+1
17	Dams displace people and submerge fertile land (1); They cause sedimentation, and silt reduces capacity (1); They affect river flow, fish migration and forests (1).	1+1+1
	OR: The movement opposed the Sardar Sarovar Dam across the Narmada (1); Tribals and farmers feared displacement and loss of livelihood (1); It raised questions of resettlement and the environment (1).	1+1+1
18	Repair leaks and avoid wastage (1); Use drip and sprinkler irrigation (1); Harvest rainwater and reuse water (1).	1+1+1
	OR: Communities build and maintain johads, tankas and kuls (1); Local rules prevent misuse (1); Their participation makes conservation effective (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Large dams are costly and cause displacement (1); Harvesting is cheap and local (1); Dams store huge amounts and generate power, but harvesting recharges groundwater (1); Dams harm ecology, harvesting does not (1); Both have roles, but harvesting is more sustainable (1).	1+1+1+1+1
	OR: Rivers flow through several states (1); Dams in one state affect water for others (1); States demand larger shares (1); Krishna-Godavari is an example of a dispute (1); These disputes need negotiated sharing (1).	1+1+1+1+1

20	The Narmada Bachao Andolan opposed the Sardar Sarovar Dam (1); Chipko-like local movements arose against Tehri dam on the Bhagirathi (1); People feared displacement and loss of livelihoods (1); They raised issues of resettlement and environment (1); Such protests forced reconsideration of dam policy (1).	1+1+1+1+1
	OR: India's population is growing and water demand is rising (1); Groundwater is being over-extracted (1); Water is polluted and unevenly distributed (1); Water is limited and needed for farming and industry (1); Conservation ensures water for future generations (1).	1+1+1+1+1

Mistakes to avoid:

- Saying water scarcity is caused only by **low rainfall**; over-use, unequal access, pollution and population growth also cause it.
- Matching projects to the wrong rivers: **Hirakud** (Mahanadi), **Bhakra-Nangal** (Sutlej-Beas), **Sardar Sarovar** (Narmada).
- Writing that Mahatma Gandhi called dams the 'temples of modern India'; the phrase is **Jawaharlal Nehru's**.
- Describing multi-purpose projects as only for **irrigation**; they also give power, flood control, water supply and navigation.
- Mixing traditional systems: **tankas** (Rajasthan), **kuls/guls** (Western Himalayas), **bamboo drip** (Meghalaya), **inundation channels** (Bengal flood plains).
- Forgetting that dams have **disadvantages** as well: sedimentation, displacement and damage to aquatic habitats.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 8: **Water Resources (Geography)** | Class 10 Social Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** Allahabad	Ganga
2	**(b)** 71%	Most is ocean
3	**(c)** collecting and storing rainwater for later use	Traditional practice
4	**(c)** compulsory in all houses	State law
5	**(c)** Sardar Sarovar dam	Displacement
6	**(b)** ice caps and glaciers	Not accessible
7	**(d)** Rajasthan	Bikaner, Phalodi
8	**(b)** temples of modern India	Progress
9	**(b)** Displacement	Problem
10	**(a)** Bikaner, Phalodi, Barmer	Arid areas

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Johads are small earthen check dams that capture and store rainwater (1); They recharge groundwater in Rajasthan (1).	1+1
12	Dams displace people and submerge forests and farmland (1); They cause sedimentation and affect river ecology (1).	1+1
13	Multipurpose river projects serve many purposes at once, such as irrigation, electricity, water supply and flood control (1); Examples are Bhakra-Nangal and Hirakud (1).	1+1
14	Kuls or guls are channels that divert water from streams for irrigation in the Western Himalayas (1); They carry glacier or spring water to fields in hills (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Tamil Nadu faced serious water shortage and falling groundwater (1); It made rooftop rainwater harvesting structures compulsory in all houses (1); Legal action against defaulters ensured its success (1).	1+1+1
16	Irrigated agriculture uses most of the water (1); Multiple cropping needs year-round irrigation (1); Green Revolution crops and tube wells increased use of groundwater (1).	1+1+1
17	Traditional methods are small, local and low-cost (1); Big dams are costly, large and cause displacement (1); Traditional methods use rain where it falls and are ecologically sound (1).	1+1+1
	OR: Multipurpose projects use dammed water for many purposes together (1); Irrigation, hydroelectric power, water supply and flood control are served (1); Examples are Bhakra-Nangal, Hirakud and Sardar Sarovar (1).	1+1+1
18	Tankas store rooftop rainwater in Bikaner, Phalodi and Barmer (1); Khadins and johads hold monsoon run-off for crops and groundwater (1); These suit the dry climate and need little cost (1).	1+1+1
	OR: Pollution makes water unfit for drinking and irrigation (1); Contaminated sources reduce usable water (1); It adds to the scarcity of clean water (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Rajasthan uses tankas, johads and khadins to store rain (1); In Himalayan areas, guls or kuls carry stream water to fields (1); Bengal has inundation channels to irrigate with floodwater (1); Meghalaya has bamboo drip irrigation (1); Shillong and Gendathur have rooftop harvesting (1).	1+1+1+1+1
	OR: Fresh water is only about 2.5 per cent of all water, and much of it is frozen (1); India has 4 per cent of the world's water and 16 per cent of its people (1); Rising population, irrigation and industry increase demand (1); Over-extraction of groundwater and pollution reduce supply (1); Unequal distribution makes scarcity worse (1).	1+1+1+1+1

20	Irrigation takes most of the water used in India (1); Multiple cropping needs more water (1); The Green Revolution increased tube well use (1); Groundwater is extracted faster than recharge (1); Water tables fall and wells run dry (1).	1+1+1+1+1
	OR: Sringaverapura near Allahabad had a floodwater harvesting system (1); In Chandragupta Maurya's time dams, lakes and irrigation systems were built (1); Sophisticated irrigation works existed in Kalinga, Nagarjunakonda, Bennur and Kolhapur (1); In the eleventh century, Bhopal Lake was one of the largest artificial lakes (1); In the fourteenth century, the Hauz Khas tank in Delhi supplied water to Siri Fort (1).	1+1+1+1+1

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